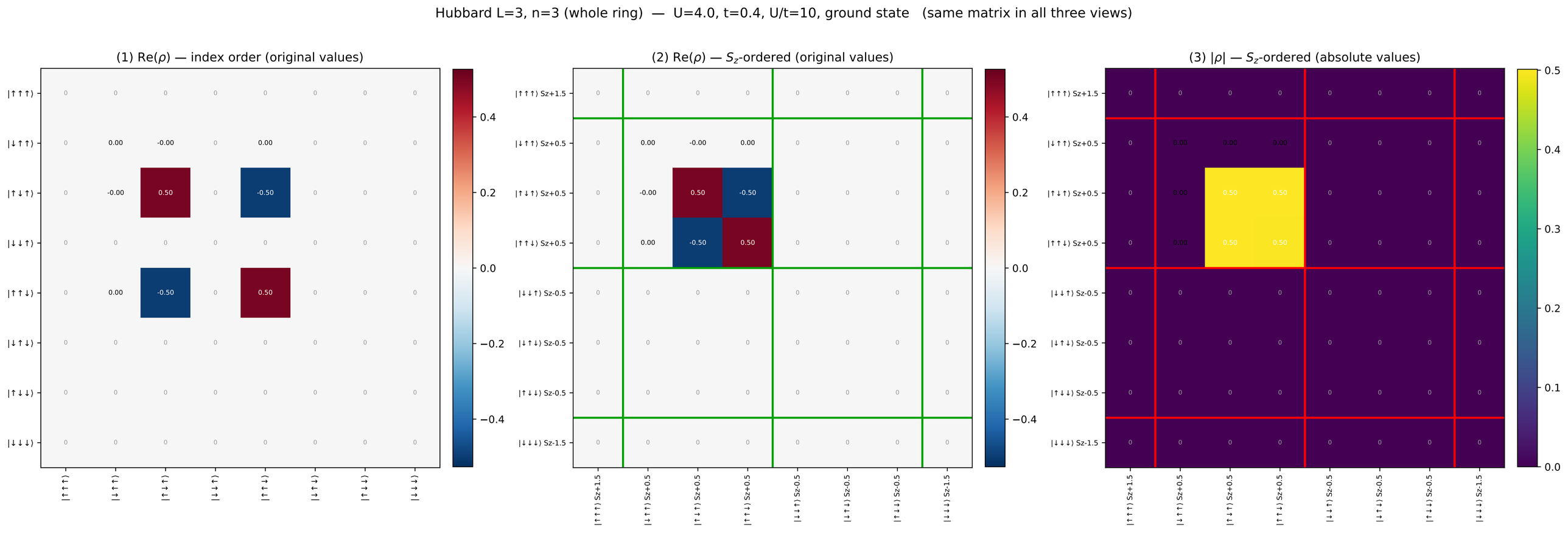
Hubbard L=3, n=3 (whole ring) — U=4.0, t=0.4, U/t=10, ground state (same matrix in all three views)
(1) Re(ρ) — index order (original values)
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑↑⟩ |↓↑↑⟩ |↑↓↑⟩ |↓↓↑⟩ |↑↑↓⟩ |↓↑↓⟩ |↑↓↓⟩ |↓↓↓⟩
|↑↑↑⟩ Sz+1.5 |↓↑↑⟩ Sz+0.5 |↑↓↑⟩ Sz+0.5 |↑↑↓⟩ Sz+0.5 |↓↓↑⟩ Sz-0.5 |↓↑↓⟩ Sz-0.5 |↑↓↓⟩ Sz-0.5 |↓↓↓⟩ Sz-1.5
0.5 0.4 0.2 0.0 −0.2 −0.4 0.1 0.3